

Chapter – 3: Electricity – Circuits and their Components

- Nihal, classmates – excited – school trip – Bhakra Nangal Dam
- Visit hydroelectric power house – force – falling water – generate electricity
- Also excited – free train ride – Nangal, Punjab to Bhakra, Himachal Pradesh – along Sutlej River, Shivalik Hills
- Before trip – students – got an assignment – prepare presentation – uses of elec.
- Looked everywhere – school, house, neighbourhood, city, internet – list kept going – organise under diff. headings
- Electricity – used all the time – let's learn in detail
- Elec. – generated – multiple ways –
 - Windmills – use wind's energy
 - Solar panels – use sun's energy
 - Falling water, natural gas, coal
- Electric supply – these src. – reach home, factories – via wires
- To learn electricity – we use – portable src. – battery
- **Caution** – danger signs – warn people – electricity – dangerous – never perform exp. with power supply at home – only use small batteries

A Torchlight

- Activity –
 - Take a torchlight – **obs.** carefully – notice – lamp, switch
 - Slide switch – obs. – lamp glows
 - Slide switch back – original position – obs. – torch lamp
- 1st position – lamp glows – other position – lamp does not glow
- Open torchlight – obs. – what's inside

A Simple Electrical Circuit

Electrical cell

- Activity –
 - Take – electric cell – look carefully
 - Notice – +ve sign, -ve sign – small metal cap – one side – flat metal disc – other side
- All electric cells – 2 terminals – one +ve – other -ve
- Metal cap – **positive terminal** – metal disc – **negative terminal**

Battery

- Activity –
 - Take – torch – 2 cells – open cell compartment – take out cells
 - Put cells – diff. order – try reversing one cell – slide switch – check lamp
 - Check order – cells placed – torch glows
- Lamp glows – cells placed – order shown in figure – notice – terminals – 2 cells – connected
- +ve terminal – 1 cell – connected to – -ve terminal – other cell
- Such combination – 2 or more cells – **battery** 
- Many devices – need more than 1 cell – connect 2 or more cells – more energy / longer time

Electric lamp

- Incandescent lamp –
 - Activity –
 - Use torch – incandescent lamp – old torchlights – confirm with teacher
 - Take – torch – **examine** lamp – notice – thin wire – middle of lamp
 - Switch on torch – notice lamp – which part glows
 - Take out lamp – help from teacher – inspect all sides
 - Thin wire – inside glass bulb – glows – called **filament**
 - Filament – attached to 2 thick wires – one connects to metal case – other connects to metal tip at base
 - These 2 terminals – fixed in such a way – do not touch each other
 - In such lamps – filament gets hot – glows – produce light
- LED lamp –
 - Many torches – used today – Light Emitting Diode (LED) lamp
 - Activity –
 - Take LED – any color – obs. – no filament inside
 - **Notice** – length – 2 wires – attached to LEDs
 - LEDs – no filament – unlike incandescent lamps
 - 2 terminals – one is longer wire (+ve) – other is shorter wire (-ve)
 - Torch – may use – one or more LEDs – sometimes – diff. shapes

Making an electric lamp glow using an electric cell or battery

- Activity –
 - Take – electric cell, incandescent lamp, cell holder, lamp holder, electric wire
 - Remove 1 cm – plastic covering – both ends – each wire – expose metal
 - Attach 2 wires – 2 ends – cell holder – insert cell – -ve terminal – spring side
 - Cell holder – not available – connect wires – directly to cells – using electrical tape
 - Attach 2 wires – screws – lamp holder – fix lamp – in the holder
 - Lamp holder – not available – connect wires – directly to cells – using electrical tape
 - Now – ready – connect cell to lamp – make it glow
 - This part of activity – 2 ways – prediction, observation
 - Connect lamp, cell – diff. combinations – shown here



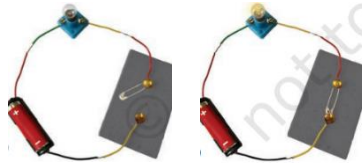
- **Predict** – each arrangement – lamp glow or not
- Now – connect – lamp, cell – obs. – lamp glows or not
- Lamp glows – arrangement – no. 1 and no. 6
- Carefully look – arrangements – lamp glows – **compare** – arrangements – lamp does not glow

An electric circuit

- Lamp glows – 1 terminal of lamp – connect to 1 terminal – elec. cell – other terminal of lamp – another terminal of cell

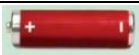
- This setup – **electrical circuit** – complete path – elec. current – flow through lamp
- Lamp glows – only – current pass-through circuit
- Dir. – elec. current – +ve to -ve terminal – elec. cell
- Terminals – lamp – connected – terminals – elec. cell – current pass – filament – incandescent lamp – it glows
- This lamp – no matter – which terminal – connect to +ve or -ve terminal – elec. cell
- Lamp glows till circuit complete – current flows through filament
- Activity –
 - Take – 2 elec. cell, LED, cell holder, 2 lengths – elec. wire
 - Remove – 1 cm – plastic covering – both ends – each wire – expose metal
 - Connect 2 wires – cell holder
 - Insert – 2 cells – holder – make sure – terminals – correct
 - Connect – free end – battery +ve terminal – longer wire – LED – other end – shorter wire – LED
 - Repeat – above step – BUT – interchange wires – connected to LED
 - LED – glows – 1st case – does not glow – 2nd case
 - Current – pass through LED – 1 dir. only
 - Current pass-through LED – only – longer wire (+ve) connected to battery +ve – AND – shorter wire (-ve) connected to battery -ve

Electric switch

- Activity –
 - Collect – 2 drawing pins, a safety pin, 2 wires, small piece of cardboard
 - Insert – drawing pin – ring – safety pin – fix it – cardboard piece – ensure – safety pin – rotate freely
 - Fix – 2nd drawing pin – cardboard piece – free end – safety pin – touch it
 - Connect wire – each drawing pin – switch is ready
 - Activity –
 - Connect – elec. cell, lamp, switch – as shown
 - Rotate – free end – safety pin – it touches – other drawing pin – as shown
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- Safety pin – touch – both drawing pins – closes gap – completes path – allows current to flow
 - This position – **ON** position – **circuit is closed** – current flows – cell's +ve to -ve terminal – lamp glows
 - Safety pin – do not touch – both drawing pins – gap in circuit – prevents current flow
 - This position – **OFF** position – **circuit is open** – lamp does not glow
 - Note – switch – place anywhere
 - **Switch** – simple device – complete or break circuit
 - Switches – light, other device – homes – work same – designed diff.

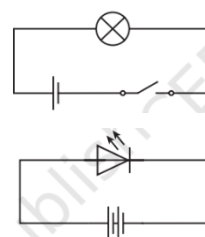
Circuit Diagrams

- Various components – elec. circuit – represent – symbols

S. No.	Electrical Component	Symbol
1.	Electric cell 	
2.	Battery 	

3.	Electric lamp 	
4.	Light Emitting Diode (LED) 	
5.	Switch – ‘ON’ position 	
6.	Switch – ‘OFF’ position 	
7.	Wire 	

- Symbol – elec. cell – longer line - +ve terminal – shorter side – -ve terminal
- Symbol – LED – triangle points to dir. – current flows – 2 arrows – indicate – light emitted
- Using symbols – represent – electrical components – easier to draw – circuit
- Representation – elec. circuit – using symbols – **circuit diagram**
- Activity –
 - Using symbols – **draw** – circuit diagram – elec. circuit



Electrical Conductors and Insulators

- Suppose – make wires – materials – not metal – use them – electrical circuit – will electric current pass?
- Activity –
 - Connect – elec. cell, lamp – leave 2 ends of wire free
 - Touch free ends – lamp glows? – yes – tester ready
 - Use tester – **identify** – materials – current pass through
 - Collect obj. – diff. materials – spoons, coins, cork, rubber, glass, keys, pins, plastic scale, wooden block, aluminium foil, candle, sewing needle, cardboard, paper, pencil lead
 - Touch free ends – tester – each obj. – make sure – wires don't touch each other
 - **Analyse** – obs. – lamp glows – which cases?
- Lamp – glows – some materials only
- Elec. current – pass through easily – some materials – not through others
- Materials – elec. current – pass through easily – **conductors**
- Materials – elec. current – does not pass – **insulators**
- Based – obs. – above act. – **conclude** – materials – conductors or insulators
- Metals – conductors – used for making wires
- Plastic, rubber, ceramics – insulators – used for covering wires
- Conductors, insulators – both imp.
- **Caution** – our body – conductor – electricity – current pass through our body – serious damage – handle electrical appliances with care